

Module 02: Agents — What Makes Something an Agent?

Course: Active Inference 101 | **Unit:** Cognitive Science | **Audience:** First-semester undergraduates

Learning Objectives

1. Distinguish an **agent** from a mere system using the concept of a generative model.
2. Explain how agents maintain a **self-model** — an internal representation of their own boundaries.
3. Describe the Active Inference view of agency: agents act to confirm their predictions about the world.
4. Identify the predict-compare-update cycle as the core process of all agents.

Introduction

Module 01 introduced systems — organized wholes separated from their environment by a Markov Blanket. But not all systems are *agents*. A whirlpool has a boundary, but it doesn't "try" to do anything. A bacterium, on the other hand, swims toward nutrients and away from toxins — it behaves *as if* it has goals. What's the difference?

Active Inference answers: an agent is a system that possesses a **generative model** — an internal model of how its sensory observations are caused by hidden states in the world. This model allows the agent to *predict*, and to *act on those predictions*.

Key Insight: You don't need a brain to be an agent. A single-celled organism has a generative model (encoded in its chemical signaling pathways) and performs active inference (swimming up chemical gradients). The difference between a bacterium and a human isn't *whether* they have a generative model — it's *how deep and complex* their model is.

Key Concepts

1. From Systems to Agents

A system becomes an agent when its internal states can be interpreted as encoding a **model of the environment**. This model has two parts:

- **Beliefs:** What the agent "thinks" is happening (its estimate of hidden states)
- **Preferences:** What the agent "wants" to happen (the sensory states it expects)

The agent then acts to make reality match its predictions. When your body "believes" it should be at 37°C and senses it's at 35°C, it shivers (acts) to bring reality in line with its model.

2. The Generative Model

A **generative model** is a probabilistic model that specifies:

- **How hidden states cause observations:** If it's raining (hidden state), I'll feel wet (observation)
- **How states change over time:** If it's raining now, it will probably still be raining in five minutes
- **What the agent prefers:** I prefer to be dry

In mathematical terms, this is written as **$p(\text{observations, states})$** — a joint probability distribution. You don't need to understand the math yet; just know that the agent has an internal "story" about how the world works.

Real-World Example — Your Morning Routine: When you wake up, your brain immediately generates predictions: "I'm in my bed, the alarm just went off, it's probably light outside." Even before opening your eyes, your generative model is making predictions. If you open your eyes and find yourself in an unfamiliar room, the resulting prediction error triggers a cascade of updating — confusion, then reorientation. The generative model is always running.

3. The Self-Model

One of the most remarkable aspects of agents is that they model *themselves*. Your brain doesn't just model the external world — it maintains a model of your own body (interoception), your position in space (proprioception), and even your own mental states (metacognition).

This self-model is what gives rise to the subjective sense of being a "self." When the self-model breaks down — as in certain neurological conditions — the sense of agency can be disrupted:

- **Alien hand syndrome:** The hand moves "on its own" — the self-model no longer claims ownership of those actions
- **Depersonalization:** "I don't feel real" — the interoceptive self-model becomes detached
- **Phantom limbs:** The self-model predicts a limb that no longer exists

4. Prediction and Prediction Error

The core process of an agent is:

1. **Predict** what sensory input should look like, based on the generative model
2. **Compare** the prediction with actual sensory input
3. **Compute prediction error:** the difference between expected and actual
4. **Minimize prediction error** by either:
5. **Updating beliefs** (perception: "Oh, it's raining") — changing the model to fit the world
6. **Acting** ("I'll open my umbrella") — changing the world to fit the model

This cycle of predict → compare → update is the heartbeat of Active Inference. It never stops — even during sleep, the brain continues to generate predictions and process errors.

5. Everyday Examples of Agency

Agent	Generative Model	Prediction	Action if Error
Bacterium	Chemical signaling pathways	"Nutrients are this direction"	Swim toward gradient
Baby	Emerging sensorimotor model	"This toy should feel like X"	Adjust grip if mismatch
Student	Knowledge of course material	"The exam will cover topics X,Y,Z"	Study what seems likely
Thermostat	Temperature setpoint	"Room should be 21°C"	Activate heater/cooler
Self-driving car	Road map + sensor model	"Lane markings should be here"	Steer to correct position

6. The Spectrum of Agency

Agency is not binary — it's a spectrum:

- **Minimal agency** (bacteria): Simple generative model, few states, limited predictions
- **Basic agency** (insects): Moderately complex model, some temporal prediction, basic learning
- **Rich agency** (mammals): Deep generative model, hierarchical predictions, flexible learning
- **Reflective agency** (humans): Meta-level model — the ability to model one's own predictions, doubt one's beliefs, and ask "am I right?"

Key Insight — Reflective Agency: What makes human cognition special isn't just the depth of our generative model — it's the fact that we can model *our own modeling*. When you realize you might be wrong about something, you are performing meta-inference — using your generative model to evaluate the

reliability of that same model. This recursive self-modeling is the basis of scientific thinking, self-doubt, and wisdom.

Summary

An agent is a system with a generative model — an internal model of how the world works, what the agent prefers, and what actions will bring observations closer to expectations. Through the predict-compare-update cycle, agents continuously minimize prediction error, either by changing their beliefs or by acting on the world. The self-model gives rise to subjective experience, and agency exists on a spectrum from bacteria to reflective human cognition.

Further Reading

- Clark, A. (2016). *Surfing Uncertainty: Prediction, Action, and the Embodied Mind*. Oxford University Press.
- Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., & Pezzulo, G. (2017). Active inference: A process theory. *Neural Computation*, 29(1), 1-49.
- Seth, A. K. (2021). *Being You: A New Science of Consciousness*. Dutton.
- Godfrey-Smith, P. (2016). *Other Minds: The Octopus, the Sea, and the Deep Origins of Consciousness*. Farrar, Straus and Giroux.